Pairing-Based SNARKs: Pinocchio and Groth16

October 29, 2026

Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book



Plan

1 Recap

2 Encrypted Verification

3 Make It Sound

4 Make it Zero-Knowledge

5 Real Protocols

Recap

Recap: R1CS

Each **constraint** in the Rank-1 Constraint System must be in the form:

$$\langle \boldsymbol{\ell}, \mathbf{z} \rangle \times \langle \boldsymbol{r}, \mathbf{z} \rangle = \langle \boldsymbol{o}, \mathbf{z} \rangle$$

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Where $\langle \mathbf{u}, \mathbf{v} \rangle$ is a dot product.

$$\langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^\top \mathbf{v} = \sum_{i=1}^{n} u_i v_i$$

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Thus

$$\left(\sum_{i=1}^{n} \ell_i z_i \right) \times \left(\sum_{j=1}^{n} r_j z_j \right) = \sum_{k=1}^{n} o_k z_k$$

That is, actually, a quadratic equation with multiple variables.

Recap: R1CS

Consider the simplest program:

```
def example(a: F, b: F, c: F) -> F:
    if a:
        return b * c
    else:
        return b + c
```

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$$x_1 \times x_1 = x_1 \quad \text{(binary check)} \tag{1}$$

$$x_2 \times x_3 = \text{mult} \tag{2}$$

$$x_1 \times \text{mult} = \text{selectMult} \tag{3}$$

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The coefficients vectors:

$$\boldsymbol{\ell}_1 = (0,0,1,0,0,0,0), \quad \boldsymbol{r}_1 = (0,0,1,0,0,0,0), \quad \boldsymbol{o}_1 = (0,0,1,0,0,0,0)$$
$$\boldsymbol{\ell}_2 = (0,0,0,1,0,0,0), \quad \boldsymbol{r}_2 = (0,0,0,0,1,0,0), \quad \boldsymbol{o}_2 = (0,0,0,0,0,1,0)$$
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$$\boldsymbol{\ell}_4 = (1,0,-1,0,0,0,0), \quad \boldsymbol{r}_4 = (0,0,0,1,1,0,0), \quad \boldsymbol{o}_4 = (0,1,0,0,0,0,-1)$$

Recap: QAP

R1CS provides us with the following constraint vectors:

$$\boldsymbol{\ell}_1, \boldsymbol{\ell}_2, \ldots, \boldsymbol{\ell}_m, \quad \boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_m, \quad \boldsymbol{o}_1, \boldsymbol{o}_2, \ldots, \boldsymbol{o}_m.$$

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Of course, they form corresponding matrices:

$$L = \begin{bmatrix} L_{1,1} & L_{1,2} & \ldots & L_{1,n} \\ L_{2,1} & L_{2,2} & \ldots & L_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ L_{m,1} & L_{m,2} & \ldots & L_{m,n} \end{bmatrix}, \text{ same goes for } R \text{ and } O$$

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An example of a single “if” statement:

$$\begin{aligned} \boldsymbol{\ell}_1 &= (0,0,1,0,0,0,0) \\ \boldsymbol{\ell}_2 &= (0,0,0,1,0,0,0) \\ \boldsymbol{\ell}_3 &= (0,0,1,0,0,0,0) \\ \boldsymbol{\ell}_4 &= (1,0,-1,0,0,0,0) \end{aligned} \qquad \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & \boxed{-1} & 0 & 0 & 0 & 0 \end{bmatrix}$$

Recap: QAP

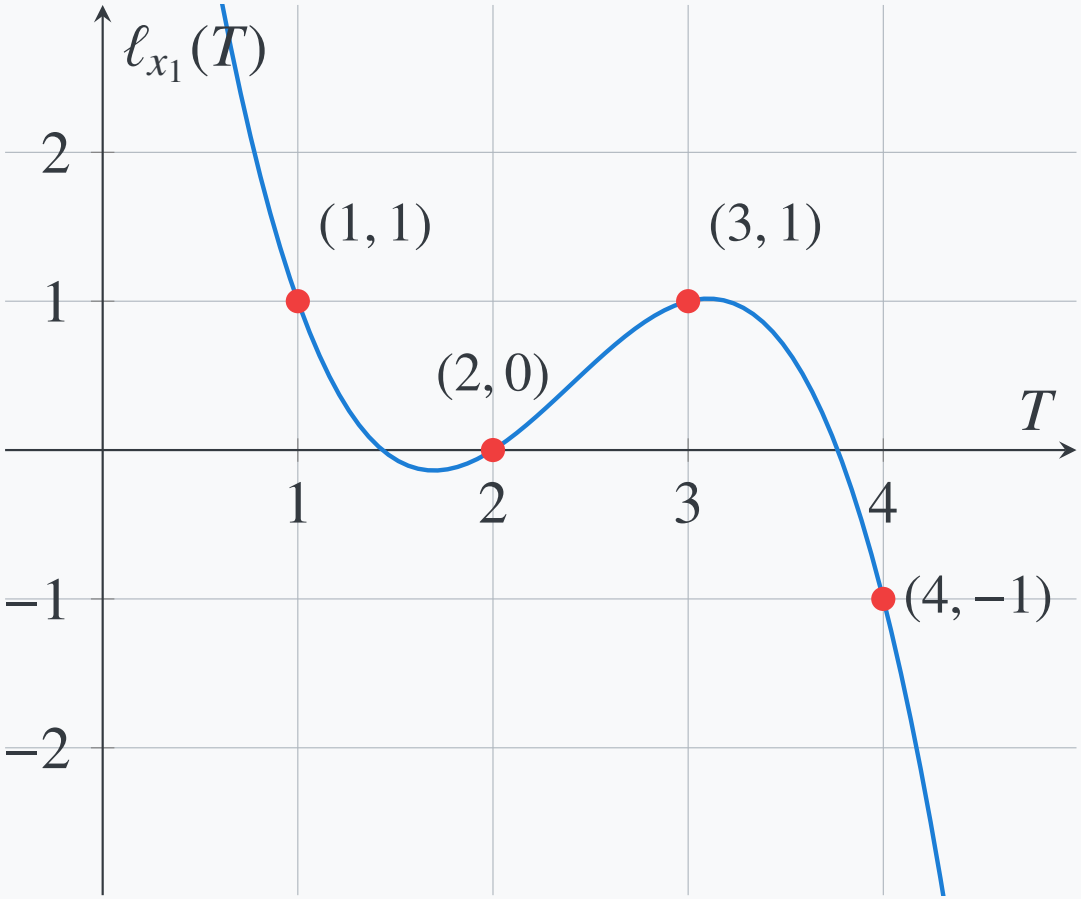


Illustration: The Lagrange interpolation polynomial for points $\{(1,1),(2,0),(3,1),(4,-1)\}$ visualized over $\mathbb{R}$.

Recap: QAP

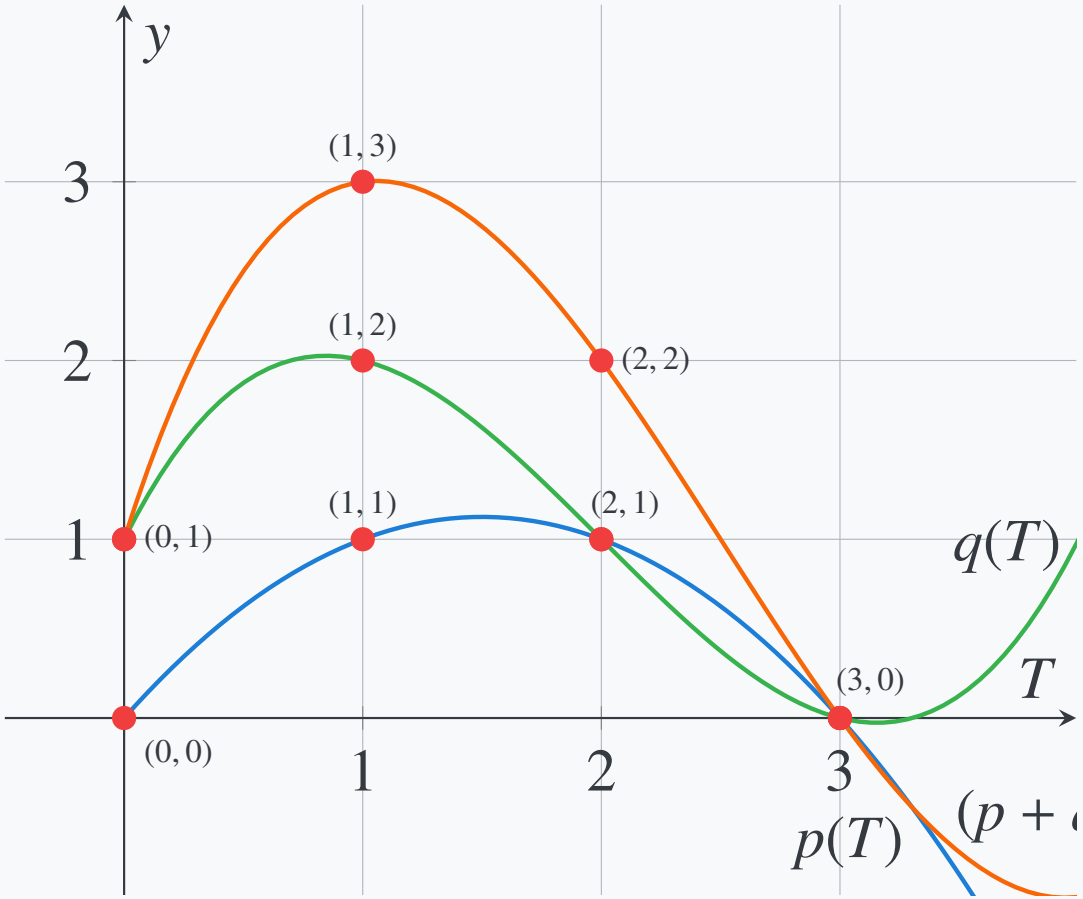


Figure: Addition of two polynomials

Recap: QAP

Now, using coefficients encoded with polynomials, we can build a constraint number $j \in \{1, \ldots, m\}$ in the next way:

$$\begin{aligned}(z_1\ell_1(j) + z_2\ell_2(j) + \cdots + z_n\ell_n(j))\times \\ \times(z_1 r_1(j) + z_2 r_2(j) + \cdots + z_n r_n(j)) = \\ =(z_1 o_1(j) + z_2 o_2(j) + \cdots + z_n o_n(j))\end{aligned}$$

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Or written more concisely:

$$\left(\sum_{i=1}^{n} z_i \ell_i(T)\right) \times \left(\sum_{i=1}^{n} z_i r_i(T)\right) = \left(\sum_{i=1}^{n} z_i o_i(T)\right)$$

$$\ell(T) \times r(T) = o(T) \quad // \text{ on } \{1, \ldots, m\}$$

Recap: QAP

Now, we can define a **master polynomial** $m(T)$, that has zeros at all elements from the set $\Omega = \{1, \ldots, m\}$

$$m(T) = \ell(T) \times r(T) - o(T)$$

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It means, that $m(T)$ can be divided by **vanishing polynomial** $z_\Omega(T)$ without a remainder!

$$z_\Omega(T) = \prod_{i=1}^{m}(T - i), \qquad h(T) = \frac{m(T)}{z_\Omega(T)} \text{ is a polynomial}$$

Encrypted Verification

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We've managed to encode into **a single polynomial** an entire computation (a program), of any size, independent of how much data it consumes.

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Where the knowledge of the correct witness is a knowledge of the quotient polynomial $h(T)$.

$$m(T) = h(T) \times z_\Omega(T)$$

Remark

Further, for brevity, we will denote $z_\Omega(T)$ as $z(T)$.

Notation Preliminaries: Groups

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Recall

The core property of the pairing function e is the **bilinearity**:

$$e(g^\alpha, g^\beta) = e(g^{\alpha\beta}, g) = e(g, g^{\alpha\beta}) = e(g, g)^{\alpha\beta}.$$

Here, g^α is the same as "scalar multiplication of a generator by a scalar $\alpha \in \mathbb{Z}_q$".

Notation Preliminaries: QAP

Recall that the core equation to be proven is:

$$\underbrace{\left(\sum_{i=1}^{n} z_i \ell_i(T)\right)}_{\text{left wires encoding}} \times \underbrace{\left(\sum_{i=1}^{n} z_i r_i(T)\right)}_{\text{right wires encoding}} - \underbrace{\left(\sum_{i=1}^{n} z_i o_i(T)\right)}_{\text{output encodings}} = z(T)h(T)$$

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Shortly, $\ell(T) \times r(T) - o(T) = z(T)h(T)$.

Naive Proof

Suppose, we are given a circuit C with a maximum degree d of polynomials used underneath.

Thus, all parties additionally know the target polynomial $z(T)$ and QAP polynomials $\{\ell_i(T)\}_{i\in[n]}, \{r_i(T)\}_{i\in[n]}, \{o_i(T)\}_{i\in[n]}$, where n is number of witness elements.

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✓ Provides witness $\mathbf{z}$ to a Verifier.

Verifier

✓ Checks $\left(\sum_{i=1}^{n} z_i \ell_i(T)\right) \times \left(\sum_{i=1}^{n} z_i r_i(T)\right) = \left(\sum_{i=1}^{n} z_i o_i(T)\right)$ on Ω.

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- ✓ Non-Interactive
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We, definitely, need to encrypt the witness data $\mathbf{z}$ somehow...

Encryption and KZG Commitment

Let us define the *encryption* operation as follows:

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Example

Consider the polynomial: $p(T) = T^2 - 5T + 2$, the encryption of $p(\tau)$ can be found as follows:

$$\text{Enc}(p(\tau)) = g^{p(\tau)} = g^{(\tau^2 - 5\tau + 2)} = \left(g^{\tau^2}\right)^1 \cdot \left(g^{\tau^1}\right)^{-5} \cdot \left(g^{\tau^0}\right)^2$$

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Question

KZG Commitment requires encrypted powers of τ: $\{g^{\tau^i}\}_{i \in [d]}$. But where can the prover take them?

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This way, we can find the KZG commitment for each polynomial. For example:

$$\mathsf{com}(\ell) \triangleq g^{\ell(\tau)} = g^{\sum_{i=0}^{d} \ell_i \tau^i} = \prod_{i=0}^{d} (g^{\tau^i})^{\ell_i},$$

Encrypted Verification

Now, we can calculate the following KZG commitments (or, synonymously, encryptions):

$$g^{\ell(\tau)}, g^{r(\tau)}, g^{o(\tau)}, g^{h(\tau)}, g^{z(\tau)}$$

But how can we verify $h(T)z(T) = \ell(T)r(T) - o(T)$ in the encrypted space?

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Well, first notice that, according to the **Schwartz-Zippel Lemma**, *with overwhelming probability* the check is equivalent to:

$$\ell(\tau)r(\tau) = z(\tau)h(\tau) + o(\tau).$$

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So, we can check this equality as follows:

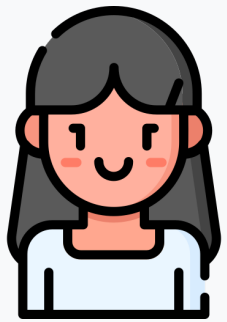
$$e(\mathrm{com}(\ell), \mathrm{com}(r)) = e(\mathrm{com}(z), \mathrm{com}(h)) \cdot e(\mathrm{com}(o), g),$$

Attempt #1: Encrypted Verification

Trusted Setup: $\tau \leftarrow\!\$\ \mathbb{F}$, $\{g^{\tau^i}\}_{i\in[d]}$, $g^{z(\tau)}$, **delete**(τ).



Prover $\mathcal{P}$



Verifier $\mathcal{V}$

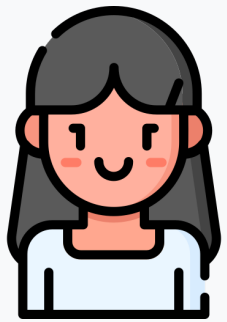
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✓ $h(T) = \frac{\ell(T)\times r(T) - o(T)}{z(T)}$.



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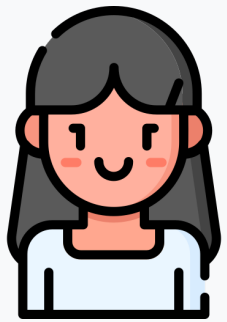
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  $\pi_L \leftarrow \mathsf{com}(\ell)$, $\pi_R \leftarrow \mathsf{com}(r)$,

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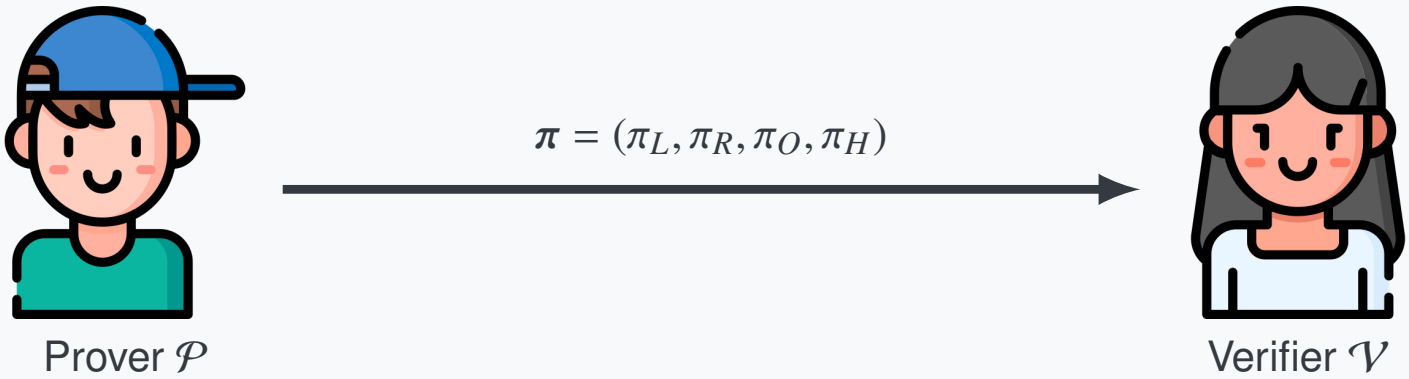
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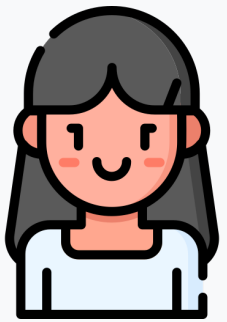
$\pi_O \leftarrow \mathsf{com}(o)$, $\pi_H \leftarrow \mathsf{com}(h)$,

✓ $e(\pi_L, \pi_R) \stackrel{?}{=} e(\mathsf{com}(z), \pi_H) \cdot e(\pi_O, g)$.



Prover $\mathcal{P}$

$\pi = (\pi_L, \pi_R, \pi_O, \pi_H)$



Verifier $\mathcal{V}$

Attempt #1: Encrypted Verification

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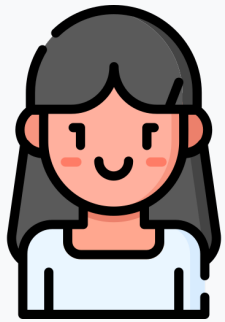
- ✗ Does it work?

Why it doesn't work??

Trusted Setup: $\tau \leftarrow\!\$\ \mathbb{F}$, $\{g^{\tau^i}\}_{i\in[d]}$, $g^{z(\tau)}$, **delete**(τ).



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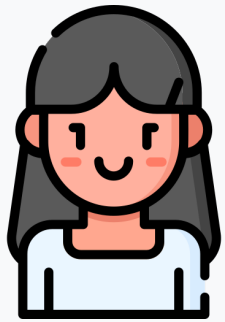
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Prover $\mathcal{P}$



Verifier $\mathcal{V}$

Problem

Prover isn't forced to use the values from the trusted setup.

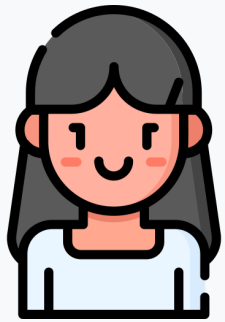
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Verifier $\mathcal{V}$

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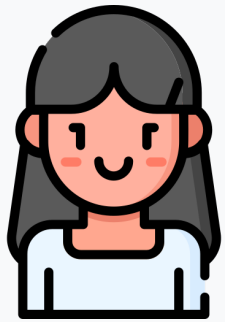
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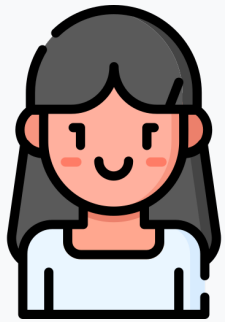
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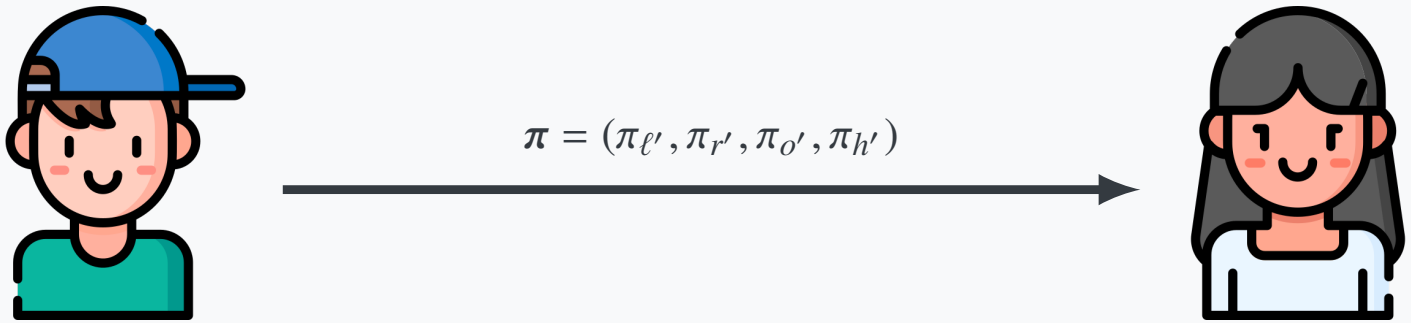
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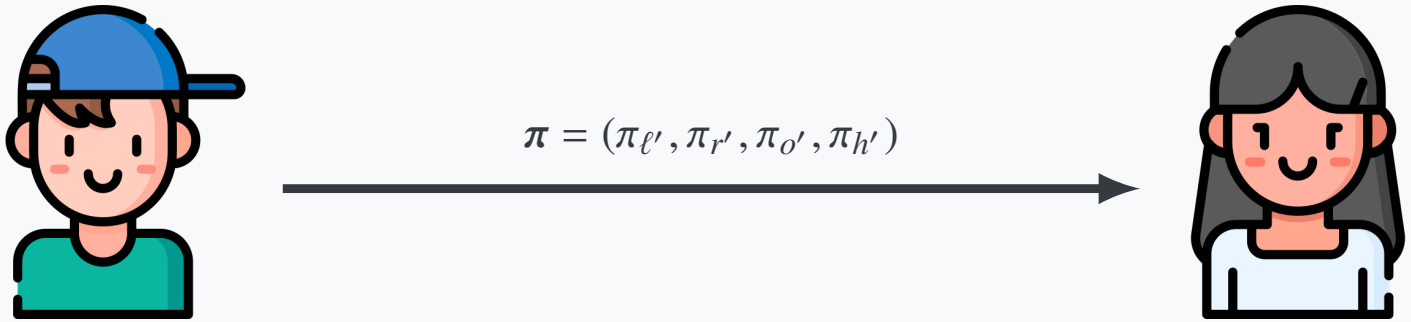
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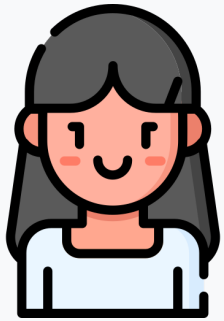
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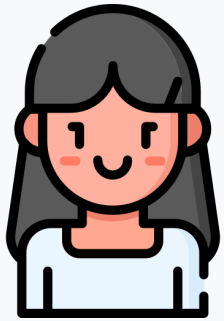
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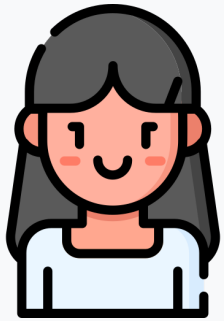
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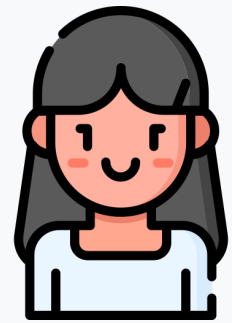
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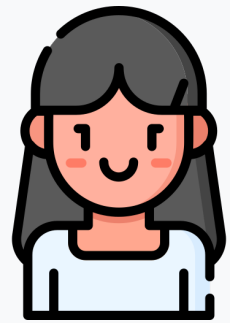
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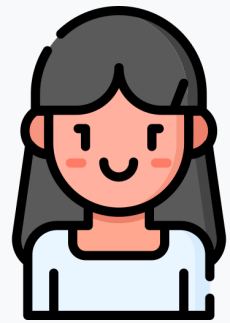
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Verifier $\mathcal{V}$

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Make It Sound

Additional Optimization

Recall that:

$$\ell(T) = \sum_{i=1}^{n} z_i \ell_i(T), \quad r(T) = \sum_{i=1}^{n} z_i r_i(T), \quad o(T) = \sum_{i=1}^{n} z_i o_i(T).$$

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Moreover, it's defined only by the circuit and trusted setup, thus, it can be calculated before proof generation as a part of the trusted setup.

Additional Optimization

Updated Trusted Setup:

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$\mathcal{P}$ can compute the KZG commitment π_L and its PoE π'_L as follows:

$$\pi_L \triangleq g^{\ell(\tau)} = g^{\sum_{i=1}^{n} z_i \ell_i(\tau)} = \prod_{i=1}^{n} (g^{\ell_i(\tau)})^{z_i},$$

$$\pi'_L \triangleq g^{\alpha\ell(\tau)} = g^{\alpha \sum_{i=1}^{n} z_i \ell_i(\tau)} = \prod_{i=1}^{n} (g^{\alpha\ell_i(\tau)})^{z_i}.$$

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To prove that the same $\mathbf{z}$ is used in all commitments, we need some "checksum" term that will somehow combine all polynomials $\ell(T)$, $r(T)$, and $o(T)$ with the witness $\mathbf{z}$.

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And we already know how to do that — PoE!

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And easy check for verifier:

$$e(\pi_L \pi_R \pi_O, g^{\beta}) = e(\pi_\beta, g).$$

One more time... that doesn't work

If the witness is consistent, the following condition must hold:

$$(z_{\ell,i}\ell_i(\tau) + z_{r,i}r_i(\tau) + z_{o,i}o_i(\tau))\beta = z_{\beta,i}\beta(\ell_i(\tau) + r_i(\tau) + o_i(\tau)) \quad \textit{// for all } i \in [n]$$

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Example

$$z_\beta = z_o = 5, \quad z_\ell = 7, \quad z_r = 3$$

$$(7 + 3)q + 5o(\tau) = 5(2q + o(\tau))$$

$$10q + 5o(\tau) = 10q + 5o(\tau)$$

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Verification:

$$e(\pi_L, g^{\beta_L}) \cdot e(\pi_R, g^{\beta_R}) \cdot e(\pi_O, g^{\beta_O}) = e(\pi_\beta, g)$$

Even more coefficients!

As the adversary has access to the public g^{β_L}, g^{β_R}, g^{β_O} he still can cheat the verifier by calculating a modified π_β.

Example

Consider a constraint $z_1 \times z_1 = z_2$. Let us try to assign 2 and 5 for z_1 in a single constraint. As $2 \times 5 = 10$, z_2 should contain the value 10.

$$\mathbf{z} = (z_1, z_2) = (2, 10)$$

Even more coefficients!

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Consider a constraint $z_1 \times z_1 = z_2$. Let us try to assign 2 and 5 for z_1 in a single constraint. As $2 \times 5 = 10$, z_2 should contain the value 10.

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Compute π_β as:

$$(g^{\beta_L \ell_1(\tau) + \beta_R r_1(\tau) + \beta_O o_1(\tau)})^2 \cdot (g^{\beta_R})^3 \cdot (g^{\beta_L \ell_2(\tau) + \beta_R r_2(\tau) + \beta_O o_2(\tau)})^{10}$$

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Proving process isn't changed, unlike verification:

$$e(\pi_L, g^{\beta_L \gamma}) \cdot e(\pi_R, g^{\beta_R \gamma}) \cdot e(\pi_O, g^{\beta_O \gamma}) = e(\pi_\beta, g^{\gamma})$$

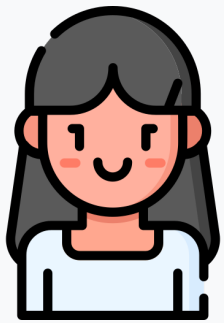
That makes it infeasible to cheat.

Attempt #3: Sound SNARK Protocol

Trusted Setup: $\tau, \alpha, \beta_L, \beta_R, \beta_O, \gamma \leftarrow\$ \mathbb{F}$, $\{\{g^{\tau^i}, g^{\alpha\tau^i}\}_{i\in[d]}$, $\{g^{\beta_L \ell_i(\tau)+\beta_R r_i(\tau)+\beta_O o_i(\tau)}\}_{i\in[n]}\}$, $\{g^{z(\tau)}, g^{\alpha}, g^{\beta_L}, g^{\beta_R}, g^{\beta_O}, g^{\beta_L\gamma}, g^{\beta_R\gamma}, g^{\beta_O\gamma}, g^{\gamma}\}$, **delete**$(\tau, \alpha, \beta_L, \beta_R, \beta_O, \gamma)$.



Prover $\mathcal{P}$



Verifier $\mathcal{V}$

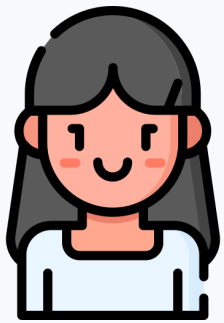
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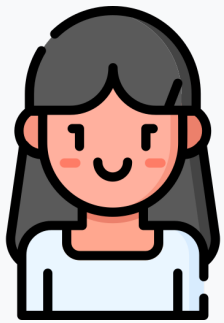
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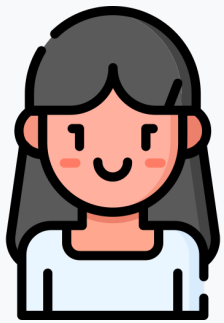
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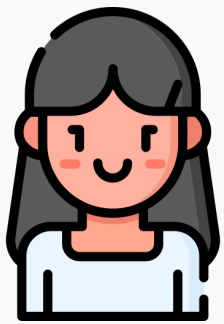
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Prover $\mathcal{P}$

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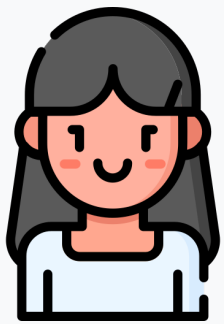
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Prover $\mathcal{P}$

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Verifier $\mathcal{V}$

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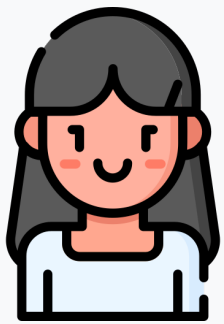
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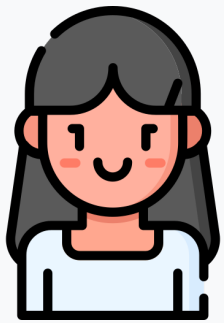
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Prover $\mathcal{P}$

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Verifier $\mathcal{V}$

Make it Zero-Knowledge

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$$\pi_L \stackrel{?}{=} \pi_R^{10}.$$

This works since $\pi_L = g^{\ell(\tau)}$ and $\pi_R = g^{r(\tau)}$, so

$$\pi_L = \pi_R^{10} \Leftrightarrow g^{\ell(\tau)} = g^{10r(\tau)} \Leftrightarrow \ell(\tau) = 10r(\tau) \Leftrightarrow \ell(T) = 10r(T)$$

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Question

So how do we represent such "noise" while preserving "homomorphic" properties of the proof?

How do we fix that? Doing stuff

Idea #1

Let prover pick random values $\delta_L, \delta_R, \delta_O, \delta_H \leftarrow\$ \mathbb{F}$ and calculate the "distorted" values:

$$\pi_L \leftarrow g^{\ell(\tau)+\delta_L}, \quad \pi_R \leftarrow g^{r(\tau)+\delta_R}, \quad \text{same for } \pi_O, \pi_H$$

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Idea #2

Let prover pick random values $\delta_L, \delta_R, \delta_O \leftarrow\$ \mathbb{F}$ and calculate:

$$\pi_L \leftarrow g^{\ell(\tau)+\delta_L z(\tau)}, \quad \pi_R \leftarrow g^{r(\tau)+\delta_R z(\tau)}, \quad \pi_O \leftarrow g^{o(\tau)+\delta_O z(\tau)}$$

Making Idea Practical

This expression can be easily evaluated. For example,

$$\pi_L = g^{\ell(\tau)} \cdot \left(g^{z(\tau)}\right)^{\delta_L} = \mathrm{com}(\ell) \cdot \mathrm{com}(z)^{\delta_L}$$

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Idea #3

Our "noise" must be controlled! This can be achieved by calculating π_H as $g^{h(\tau)+\Delta_H}$ where Δ_H depends on $\delta_L, \delta_R, \delta_O$ (and possibly other public parameters).

Making Idea Practical

Check $e(\pi_L, \pi_R) = e(\text{com}(z), \pi_H) \cdot e(\pi_O, g)$ is now equivalent to:

$$(\ell(T) + \delta_L z(T))(r(T) + \delta_R z(T)) = (h(T) + \Delta_H)z(T) + (o(T) + \delta_O z(T)),$$

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which, by expanding, gives us the following equation:

$$\begin{aligned} \cancel{\ell(T)r(T)} + \delta_R \ell(T)z(T) + \delta_L z(T)r(T) + \delta_L \delta_R z(T)^2 \\ = \cancel{h(T)z(T) + o(T)} + \Delta_H z(T) + \delta_O z(T) \end{aligned}$$

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$$(\ell(T) + \delta_L z(T))(r(T) + \delta_R z(T)) = (h(T) + \Delta_H)z(T) + (o(T) + \delta_O z(T)),$$

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$$\begin{aligned} \cancel{\ell(T)r(T)} + \delta_R \ell(T)z(T) + \delta_L z(T)r(T) + \delta_L \delta_R z(T)^2 \\ = \cancel{h(T)z(T) + o(T)} + \Delta_H z(T) + \delta_O z(T) \end{aligned}$$

where we can cancel out $\ell(T)r(T)$ and $h(T)z(T) + o(T)$ terms since they are equal based on initial construction. This way, we get the following expression for Δ_H:

$$\boxed{\Delta_H = \delta_O + \delta_R \ell(T) + \delta_L r(T) + \delta_L \delta_R z(T)}$$

Witness Consistency Proof

Finally, let us not forget about π_β! Previously, we had:

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Therefore, our new π_β becomes:

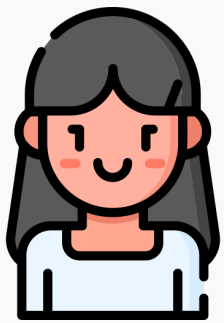
$$\pi_\beta = \left(g^{\beta_L z(\tau)}\right)^{\delta_L} \left(g^{\beta_R z(\tau)}\right)^{\delta_R} \left(g^{\beta_O z(\tau)}\right)^{\delta_O} g^{\beta_L \ell(\tau) + \beta_R r(\tau) + \beta_O o(\tau)}$$

Overall Protocol

Trusted Setup: $\tau, \alpha, \beta_L, \beta_R, \beta_O, \gamma \leftarrow\!\$\ \mathbb{F}$, $\{\{g^{\tau^i}, g^{\alpha\tau^i}\}_{i\in[d]}$, $\{g^{\beta_L \ell_i(\tau)}, g^{\beta_R r_i(\tau)}, g^{\beta_O o_i(\tau)}\}_{i\in[n]}\}$, $\{g^{z(\tau)}, g^{\alpha}, g^{\beta_L}, g^{\beta_R}, g^{\beta_O}, g^{\beta_L\gamma}, g^{\beta_R\gamma}, g^{\beta_O\gamma}, g^{\gamma}\}$, **delete**$(\tau, \alpha, \beta_L, \beta_R, \beta_O, \gamma)$.



Prover $\mathcal{P}$



Verifier $\mathcal{V}$

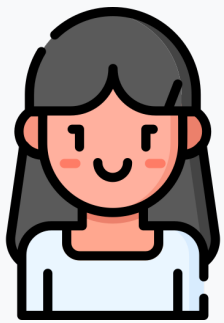
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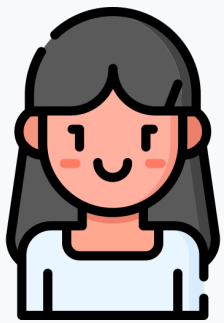
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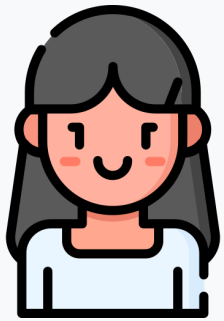
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Prover $\mathcal{P}$



Verifier $\mathcal{V}$

Overall Protocol

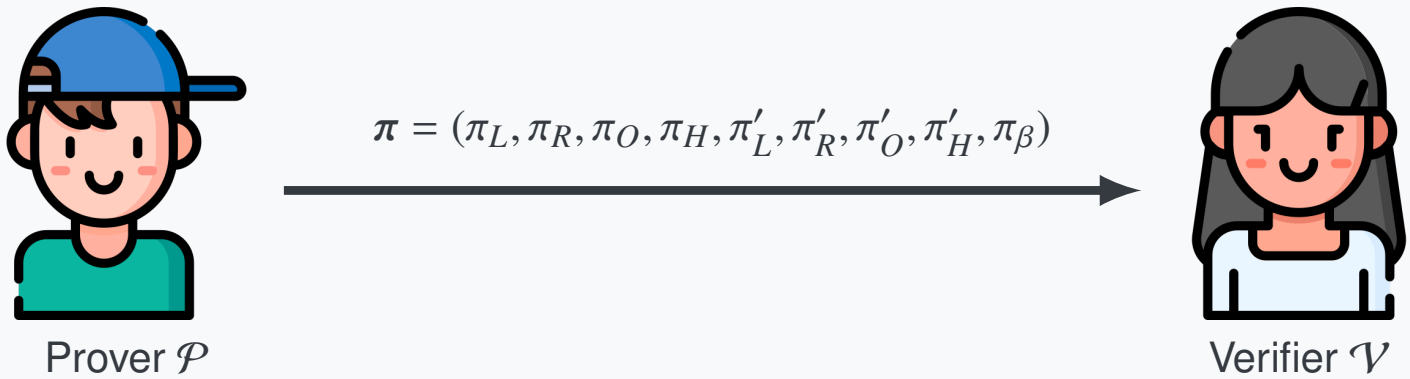
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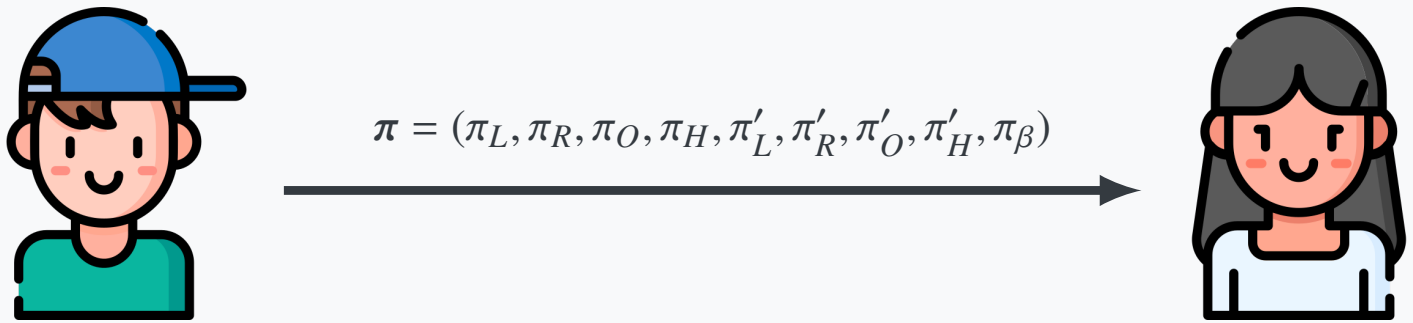
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✓ $e(\pi_L, \pi_R) \stackrel{?}{=} e(\mathsf{com}(z), \pi_H) \cdot e(\pi_O, g)$.

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Overall Protocol

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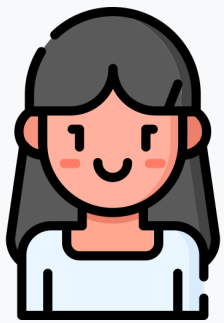
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Prover $\mathcal{P}$

$$\boldsymbol{\pi} = (\pi_L, \pi_R, \pi_O, \pi_H, \pi'_L, \pi'_R, \pi'_O, \pi'_H, \pi_\beta)$$



Verifier $\mathcal{V}$

Overall Protocol

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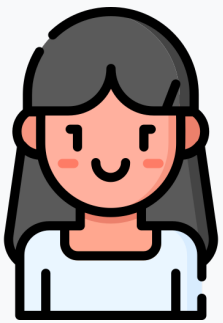
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- ✓ $e(\pi_L, g^{\gamma\beta_L}) \cdot e(\pi_R, g^{\gamma\beta_R}) \cdot e(\pi_O, g^{\gamma\beta_O}) = e(\pi_\beta, g^\gamma)$



Prover $\mathcal{P}$

$$\boldsymbol{\pi} = (\pi_L, \pi_R, \pi_O, \pi_H, \pi'_L, \pi'_R, \pi'_O, \pi'_H, \pi_\beta)$$



Verifier $\mathcal{V}$

Real Protocols

Public and Private Inputs

So far, the verifier did not use the public statement at all. Split the indices of the extended witness $\mathbf{z}$ into $\mathcal{I}_{\text{io}}$ (input/output, public) and $\mathcal{I}_{\text{mid}}$ (intermediate, private), with $\mathcal{I}_{\text{mid}} \cap \mathcal{I}_{\text{io}} = \emptyset$, $\mathcal{I}_{\text{mid}} \cup \mathcal{I}_{\text{io}} = \{1, \ldots, n\}$.

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$$\pi_L = \underbrace{g^{\sum_{i \in \mathcal{I}_{\text{mid}}} z_i \ell_i(\tau)}}_{\text{new } \pi_L} \times \underbrace{g^{\sum_{i \in \mathcal{I}_{\text{io}}} z_i \ell_i(\tau)}}_{\pi_{\ell,\text{io}}}.$$

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Remark

The prover computes π_L, π_R, π_O only over $\mathcal{I}_{\text{mid}}$, while the verifier restores $\pi_L^* \leftarrow \pi_L \prod_{i \in \mathcal{I}_{\text{io}}} (g^{\ell_i(\tau)})^{z_i}$ (and similarly π_R^*, π_O^*) from the public values and uses them in the polynomial equality check.

Complexity of the Basic Protocol

Proposition (Complexity of the Basic Protocol)

Suppose circuit consists of n gates. Then, the complexity of the basic protocol is as follows:

- **Proof Size:** $O(1)$ — constant number of group elements.
- **Setup Time:** $O(n)$ — calculating powers of τ, evaluations at τ.
- **Prover Time:** $O(n \log n)$ — using FFT and wise choice of Ω.
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We can do better!

Pinocchio Protocol

Idea

In toxic waste, include $\rho_L, \rho_R \leftarrow\$ \mathbb{F}$, set $\rho_O \leftarrow \rho_L \rho_R$, and define the following generators:

$$g_L \leftarrow g^{\rho_L}, \quad g_R \leftarrow g^{\rho_R}, \quad g_O \leftarrow g^{\rho_O}$$

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Reason

Such choice of generators reduces 15 pairings to **11 pairings**. Additionally, we have only **8 group elements** in the proof.

Groth16 Protocol

Idea: Generic Group Model

Use **Generic Group Model** (GGM) technique. Simply put, GGM allows the adversary to only make **oracle requests** to compute the group operations. For example, having a set $\{g^{\alpha r_i(\tau)}\}_{i\in[d]}$, adversary can compute only linear combinations of these values. In the particular case of Groth16, instead of considering $\ell_i(T)$, $r_i(T)$, and $o_i(T)$ separately, we construct their linear combinations as $Q_i(T) := \beta\ell_i(T) + \alpha r_i(T) + o_i(T)$, where α and β are toxic parameters.

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💬 *Asymmetric Pairing*

Groth16 uses the asymmetric pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ with generators $g_1 \in \mathbb{G}_1$ and $g_2 \in \mathbb{G}_2$: it is more efficient than the symmetric one.

Groth16 Protocol: Setup Procedure

Sample $\alpha, \beta, \gamma, \delta, \tau \leftarrow\$ \mathbb{F}$. The proving key is formed as follows:

$$\mathsf{pp} \leftarrow \Big(g_1^{\alpha}, g_1^{\beta}, g_1^{\delta}, \Big\{g_1^{\tau^i}, g_1^{(\beta\ell_i(\tau)+\alpha r_i(\tau)+o_i(\tau))/\gamma}, g_1^{\tau^i z(\tau)/\delta}\Big\}_{i\in[n]},$$
$$g_2^{\beta}, g_2^{\delta}, g_2^{\gamma}, \{g_2^{\tau^i}\}_{i\in[d]}\Big)$$

Groth16 Protocol: Proving Procedure

Sample random $\delta_L, \delta_R \leftarrow\$ \mathbb{F}$ and compute the following values:

$$\pi_L \leftarrow g_1^{\alpha+\sum_{i=1}^{n} z_i \ell_i(\tau)+\delta_L \delta}, \quad \pi_R \leftarrow g_2^{\beta+\sum_{i=1}^{n} z_i r_i(\tau)+\delta_R \delta},$$

$$\pi_O \leftarrow g_1^{\frac{Q_{\mathsf{mid}}(\tau)+h(\tau)z(\tau)}{\delta}+\ell\delta_R+r\delta_L-\delta_L\delta_R\delta},$$

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where by Q_{mid} we denoted the following expression:

$$Q_{\mathsf{mid}}(\tau) = \sum_{i \in \mathcal{I}_{\mathsf{mid}}} z_i Q_i(\tau) = \sum_{i \in \mathcal{I}_{\mathsf{mid}}} z_i(\beta \ell_i(\tau) + \alpha r_i(\tau) + o_i(\tau))$$

Groth16 Protocol: Verification Procedure

The verifier first calculates the following value:

$$\pi_{\mathsf{io}} \leftarrow g_1^{\sum_{i \in \mathcal{I}_{\mathsf{io}}} z_i(\beta \ell_i(\tau) + \alpha r_i(\tau) + o_i(\tau))/\gamma},$$

Groth16 Protocol: Verification Procedure

The verifier first calculates the following value:

$$\pi_{\mathsf{io}} \leftarrow g_1^{\sum_{i \in \mathcal{I}_{\mathsf{io}}} z_i(\beta \ell_i(\tau) + \alpha r_i(\tau) + o_i(\tau))/\gamma},$$

and then checks the following single condition:

$$e(\pi_L, \pi_R) = e(g_1^{\alpha}, g_2^{\beta}) e(\pi_{\mathsf{io}}, g_2^{\gamma}) e(\pi_O, g_2^{\delta})$$

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Note

$e(g_1^{\alpha}, g_2^{\beta})$ can be additionally hard-coded in the verifier, thus reducing the number of pairings to 3. Finally, the proof's size is now reduced to 3 group elements: two from $\mathbb{G}_1$, and one from $\mathbb{G}_2$.

Thank you for your attention



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